

hw9

1. lets say n is a positive integer then we use induction to prove the following:

$$1 + 5 + 9 + \cdots + (4n - 3) = 2n^2 - n$$

Proof. so we use ordinary induction on $n \in \mathbb{Z}_{>0}$

base case: $n = 1$ so the left-hand side is just 1, and the right-hand side is

$$2(1)^2 - 1 = 1$$

so the statement holds for $n = 1$

Inductive step: let's assume that for some positive integer k ,

$$1 + 5 + 9 + \cdots + (4k - 3) = 2k^2 - k$$

we have to to prove that

$$1 + 5 + 9 + \cdots + (4k - 3) + (4(k + 1) - 3) = 2(k + 1)^2 - (k + 1)$$

starting from the left side and using the inductive hypothesis,

$$\begin{aligned} 1 + 5 + 9 + \cdots + (4k - 3) + (4(k + 1) - 3) &= (2k^2 - k) + (4k + 1) \\ &= 2k^2 + 3k + 1 \end{aligned}$$

s now simplify the desired right-hand side:

$$2(k + 1)^2 - (k + 1) = 2(k^2 + 2k + 1) - k - 1 = 2k^2 + 3k + 1$$

so,

$$1 + 5 + 9 + \cdots + (4(k + 1) - 3) = 2(k + 1)^2 - (k + 1)$$

so the statement holds for $k + 1$

therefore, by mathematical induction,

$$1 + 5 + 9 + \cdots + (4n - 3) = 2n^2 - n$$

for all positive integers n

□

2. so we let n be a positive integer then we prove by smallest counter example that:

$$\forall n > 0 \quad \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

Proof. we argue by smallest counterexample

so we assume for contradiction sake that the statement is false and then there exists at least one positive integer n such that

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} \neq \frac{2^n - 1}{2^n}$$

we let m be the smallest positive integer for which the equality fails

since m is the smallest counterexample, $m \neq 1$ because when $n = 1$ we have

$$\frac{1}{2} = \frac{2^1 - 1}{2^1} = \frac{1}{2}$$

so $m > 1$

by minimality of m , the statement must be true for $m - 1$ so then,

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^{m-1}} = \frac{2^{m-1} - 1}{2^{m-1}}$$

now I add $\frac{1}{2^m}$ to both sides:

$$\begin{aligned} \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^{m-1}} + \frac{1}{2^m} &= \frac{2^{m-1} - 1}{2^{m-1}} + \frac{1}{2^m} \\ &= \frac{2(2^{m-1} - 1)}{2^m} + \frac{1}{2^m} \\ &= \frac{2^m - 2 + 1}{2^m} \\ &= \frac{2^m - 1}{2^m} \end{aligned}$$

so the formula is true for $n = m$, contradicting the fact that m was a counterexample

so this contradiction pretty much shows that no counterexample exists so,

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

for every positive integer n

□

3.

so there exist non-negative integers x, y such that $n = 3x + 7y$ for all $n \geq 12$

Proof. first we let $P(n)$ be the statement:

“there exist nonnegative integers x, y such that $n = 3x + 7y$.”

we’ll prove $P(n)$ for all $n \geq 12$ by strong induction

base cases:

$$12 = 3 \cdot 4 + 7 \cdot 0,$$

$$13 = 3 \cdot 2 + 7 \cdot 1,$$

$$14 = 3 \cdot 0 + 7 \cdot 2$$

so $P(12)$ and $P(13)$ and $P(14)$ are all true

inductive step: we fix some integer $k \geq 14$ and assume that $P(j)$ is true for every integer j with

$$12 \leq j \leq k$$

we have to prove $P(k+1)$

because $k \geq 14$, we have $k-2 \geq 12$ and by the strong inductive hypothesis, $P(k-2)$ is true so therefore there exist nonnegative integers x, y such that

$$k-2 = 3x + 7y$$

add 3 to both sides:

$$k+1 = (k-2) + 3 = 3x + 7y + 3 = 3(x+1) + 7y$$

since $x+1$ and y are still nonnegative integers, this shows that $P(k+1)$ is true

so therefore, by strong induction, for every integer $n \geq 12$ there exist non-negative integers x, y so that

$$n = 3x + 7y$$

□

4.

Proof. we'll enumerate all ordered pairs of integers
for each integer $k \geq 0$, define the finite set

$$S_k = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : \max(|a|, |b|) = k\}$$

so:

- $S_0 = \{(0, 0)\}$,
- S_1 consists of all pairs on the square of "radius" 1,
- S_2 consists of all pairs on the square of "radius" 2,
- and so on

each S_k is finite and $|S_0| = 1$ and $|S_k| = 8k$ for $k \geq 1$

so now we enumerate the pairs shell by shell:

$$S_0, S_1, S_2, S_3, \dots$$

in each shell S_k (for $k \geq 1$) we list the elements in the this order:

$$\begin{aligned} &(k, -k), (k, -k+1), \dots, (k, k), \\ &(k-1, k), (k-2, k), \dots, (-k, k), \\ &(-k, k-1), (-k, k-2), \dots, (-k, -k), \\ &(-k+1, -k), (-k+2, -k), \dots, (k-1, -k). \end{aligned}$$

this lists every point on the boundary of the square once

so the full enumeration is

$$\begin{aligned} &(0, 0), \\ &(1, -1), (1, 0), (1, 1), (0, 1), (-1, 1), (-1, 0), (-1, -1), (0, -1), \\ &(2, -2), (2, -1), (2, 0), (2, 1), (2, 2), (1, 2), (0, 2), (-1, 2), (-2, 2), \\ &(-2, 1), (-2, 0), (-2, -1), (-2, -2), (-1, -2), (0, -2), (1, -2), \\ &\dots \end{aligned}$$

this is a sequence indexed by the positive integers, so every pair $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ belongs to one shell S_k , $k = \max(|a|, |b|)$, and then appears once in the listing of that shell, so every element of $\mathbb{Z} \times \mathbb{Z}$ appears once in our sequence

so there's a bijection between $\mathbb{N}$ and $\mathbb{Z} \times \mathbb{Z}$, so $\mathbb{Z} \times \mathbb{Z}$ is countable □